



Cape Peninsula
University of Technology

FACULTY OF INFORMATICS & DESIGN

Diploma in Information & Communication Technology

Group Report

Project 3 – PRT362S

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Introduction

During this difficult period students are forced to work from home, and as a result the deliverables for their various subjects and projects start to become difficult to manage. We aim to tackle this problem by giving students the initiative to keep track of their work and when things are expected of them. Our aim is to create an “Assignment Tracker”, this application will help students manage their assignments, by keeping track of deadlines and managing the various tasks that are associated with their assignment deliverables. Currently, students and lecturers are relegated to using Blackboard as their primary source of interaction with their course work, we feel that it puts too much of the responsibility on the lecturer to manage all the work expected of their students, the aim of this application is to bridge that gap and allow students to take some of that responsibility and be in the driving seat when managing their work.

Our group has been working on the project for the past 6 months and have to come to conclusion as to how to create a solution. The project aims at producing a web application that allows students at the Cape Peninsula University of Technology to track their assignments as well as manage their individual and group assignments with tools that the platform will provide. This document will discuss the planning and design of the group’s proposed solution.

Design

Home Page

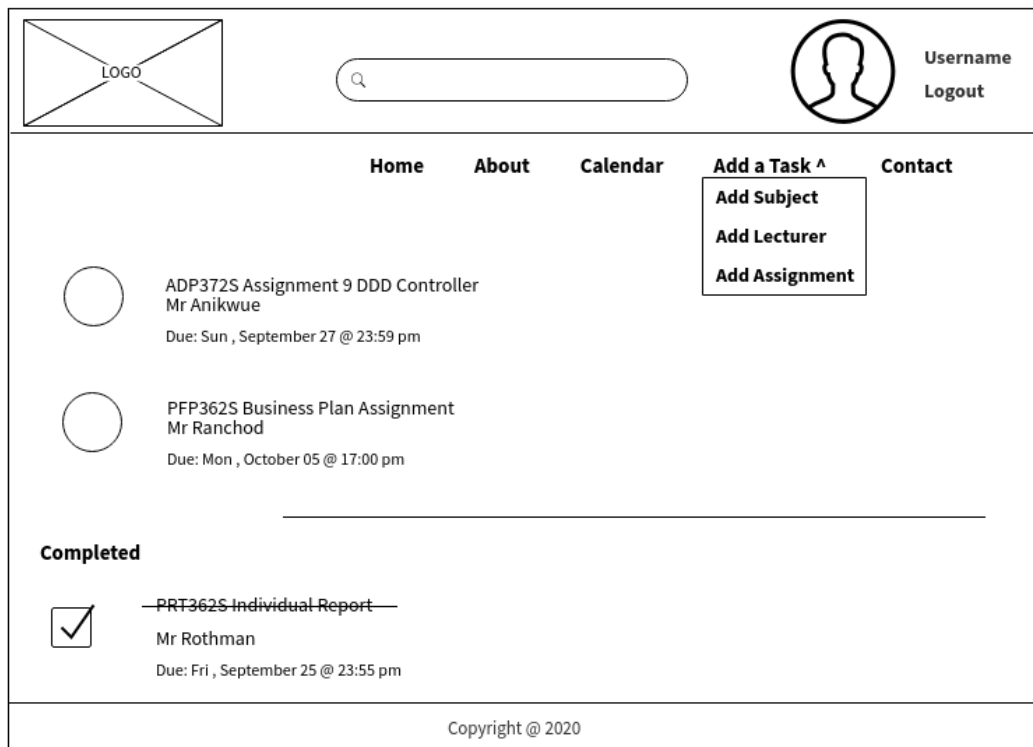


Figure 1 shows the landing page of the website, with all the services that the website offers laid in front of the user to explore

The home page is the landing page after the user has logged in or registered. The page also serves as a holder for the rest of the websites' pages. What I mean by this is that in order to access the pages such as contact, about, calendar...etc, one needs to navigate on this page and choose the desired option/page to land on the required page.

Design Tools

WordPress

Elementor - frontend drag & drop page builder

Registration Page

Header
Register An Account! User Email * Confirm Email * First Name * Last Name * Student Number * Password * Confirm Password * <input type="submit" value="Submit"/>

Figure 2 shows the details required should the user want to create an account on the website

When a student first visits the website they should be able to view the services that the website offers, however they should not be able to interact with these services until they registered an account.

Design Tools

WordPress

Elementor - frontend drag & drop page builder

User Registration - drag and drop user registration form and login form builder

Login Page

Header
Login To Your Account User Email * Password * Submit

Figure 3 shows the login information required from the user once they have created an account on the website

The login page will use the student's information entered during the registration phase to sign in to the website.

Design Tools

WordPress

Elementor - frontend drag & drop page builder

User Registration - drag and drop user registration form and login form builder

Assignment Creation

Header

 Username
Log out

Create A New Assignment

Date assigned



Deadline



☒ Individual Assignment

☐ Group Assignment

Save

Figure 4 shows the user interface to create new assignments

This component manages how assignments are created, updated and to whom they are accessible. Users should only be allowed access to this service after they have created an account.

Design Tools

WordPress

Elementor - frontend drag & drop page builder

Assignment Editing

Header

Username
Log out

Edit Assignment

Project Delievrable 4

Project 3

Date assigned

12 May 2020

Deadline

25 September 2020

☐ Individual Assignment

☒ Group Assignment

Invite New Members

View All Members

UPDATE

Figure 5 shows the user interface to edit assignments

Viewing All Assignments

Header

Username
Log out

All Assignment

Due Today

320x170

Project Delievable 3

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nunc maximus, nulla ut commodo sagittis, sapien dui mattis dui, non pulvinar lorem felis nec erat.

Contribute

Delete

320x170

ADP 3

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nunc maximus, nulla ut commodo sagittis, sapien dui mattis dui, non pulvinar lorem felis nec erat.

Contribute

Delete

OverDue

320x170

Assignment Name

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Contribute

Delete

Upcoming

320x170

Assignment Name

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Contribute

Delete

320x170

Assignment Name

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Contribute

Delete

Submitted

320x170

Assignment Name

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nunc maximus, nulla ut commodo sagittis, sapien dui mattis dui, non pulvinar lorem felis nec erat.

Contribute

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Assignment Name

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nunc maximus, nulla ut commodo sagittis, sapien dui mattis dui, non pulvinar lorem felis nec erat.

Contribute

Delete

320x170

Assignment Name

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nunc maximus, nulla ut commodo sagittis, sapien dui mattis dui, non pulvinar lorem felis nec erat.

Contribute

Delete

Figure 6 shows the user interface to view assignments

Add Lecturer Page

The screenshot shows a web application interface for adding a new lecturer. At the top, there is a dark grey header bar. On the left of the header is a 'Logo' placeholder. On the right, there is a user profile icon with the text 'User name' and 'Logout' below it. In the center of the header is a navigation menu with links: 'Add Assignment', 'Add subject', 'Add lectures', 'Calender', 'About', and 'Contact'. Below the header, the main content area has a title 'Add new lecturer' centered at the top. On the left side of this area, there is a vertical sidebar with three buttons: 'Add New Lecturer', 'Assign New Task', and 'Track Assignment'. The central part of the main area contains a form with the following fields: a dropdown menu for 'Mr' (with a downward arrow), a text input for 'Name', a text input for 'Surname', a text input for 'Email', a text input for 'Cell Number', and a dropdown menu for 'Select Subject' (with a downward arrow). Below these fields is a dark grey button labeled 'Add'. At the bottom of the page is a dark grey footer bar with the word 'Footer' centered.

Figure 7 show the add lecturer webpage design

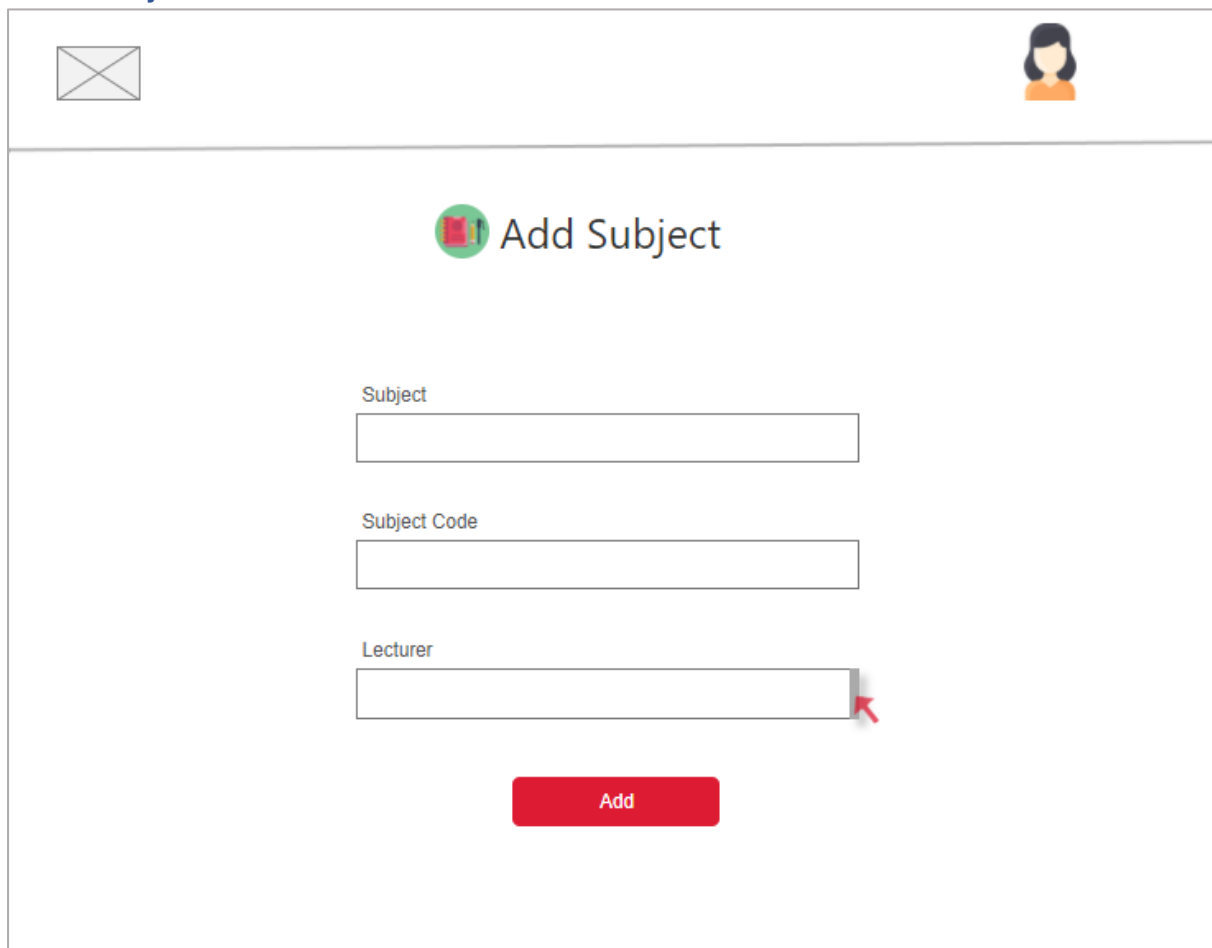
This component allows lecturers to register to the website or log in if they're already registered. This component aims to make it easier for lecturers to find the information needed to manage all the work expected of their students. Lecturers should be able to communicate with students that are associated with their class.

Design Tools

WordPress

Elementor - frontend drag & drop page builder

Add Subject



The wireframe shows a web page titled "Add Subject". At the top left is a placeholder for a logo (a square with an 'X'). At the top right is a placeholder for a user profile picture (a person icon). The main content area has a green circular icon with a book symbol next to the title "Add Subject". Below the title are three text input fields: "Subject", "Subject Code", and "Lecturer". A red arrow points to the right side of the "Lecturer" field. At the bottom center is a red button labeled "Add".

Figure 8 shows the add subject wireframe

The “add subject” webpage module allows a registered user to add subjects, subject codes and assign existing lecturers.

The module works with the “Lectures” module as a user will assign an existing lecturer, this requires for a lecturer to already exist. When user attempts to click the “add” button without filling in the required fields, a message requesting to do so will be presented to the user.

Design Tools

WordPress

Elementor - frontend drag & drop page builder

Calendar

The wireframe shows a web application interface for a calendar. At the top, there is a browser address bar with the URL `http://assignmenttrackerapp.azurewebsites.net/duedates`. Below this is a large rectangular area with a diagonal cross, likely a placeholder for a logo or image. Underneath is a search bar with the placeholder text "Search for assignment" and a blue "FIND" button. To the right of the search bar are links for "List", "Month", and "Day" views. The main content area is divided into two sections. The left section displays a list of assignments. The first entry is for July 19, 2020, at 10:00 am, titled "ADP372S Practical 1 - Inheritance". The second entry is for August 24, 2020, at 23:59 pm, titled "ITS372S Exercise- SQL". The right section is titled "CALENDAR" and shows a calendar for January 2017. The calendar grid has columns for days of the week (Mo, Tu, We, Th, Fr, Sa, Su) and rows for dates (1-31). A blue "Add Due Date" button is located at the bottom right of the calendar grid.

http://assignmenttrackerapp.azurewebsites.net/duedates

Search for assignment **FIND** [List](#) | [Month](#) | [Day](#)

< > Today **Now - 2020 - 09 - 11**

CALENDAR

July 2020

MON

19 **ADP372S**
Practical 1 - Inheritance

August 2020

WED

24 **ITS372S**
Exercise- SQL

January 2017

Mo	Tu	We	Th	Fr	Sa	Su
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31	Add Due Date			

Figure 9 shows the calendar design wireframe

This page will make use of the Google calendar plugin to link your Google calendar to the website. This will allow users to view assignments and tasks due in an easier-to-view calendar format. The students will also be allowed to assign assignments to dates from this screen.

Design Tools

WordPress

Elementor - frontend drag & drop page builder

Google Calendar API

About Page

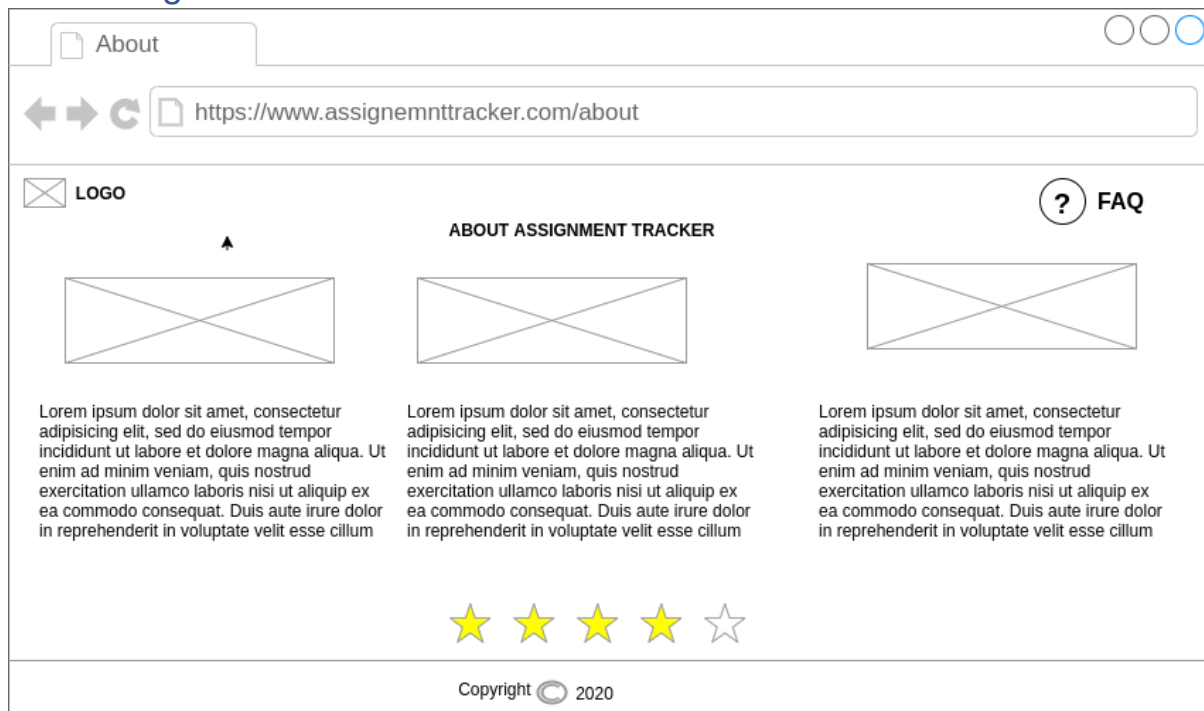


Figure 10 shows the wireframe of the About Page

In order to access About page, the user must land on Home page via the URL and then click About. Figure 1 is the wireframe of about page that the user will see.

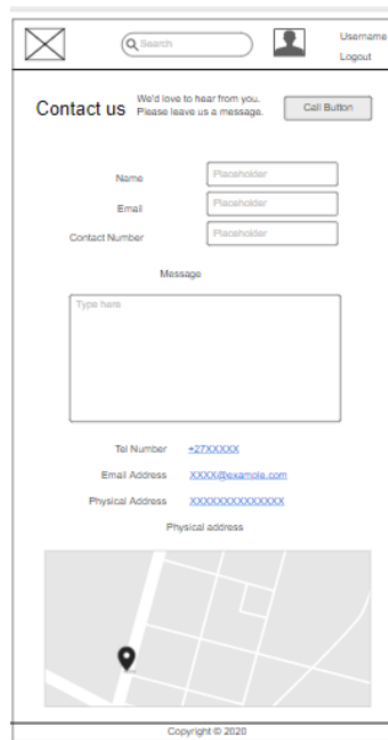
As seen in figure one, there is a Frequently Asked Questions button available for the user to get their unanswered concerns cleared. Figure 2 illustrates the outcome of clicking FAQ.

Design Tools

WordPress

Elementor - frontend drag & drop page builder

Contact Page



The wireframe shows a contact page layout. At the top, there is a navigation bar with a close icon, a search bar, a user profile icon, and links for 'Username' and 'Logout'. Below this, the main content area is titled 'Contact us' with a sub-header 'We'd love to hear from you. Please leave us a message.' and a 'Call Button'. The form includes input fields for 'Name', 'Email', and 'Contact Number', each with a 'Placeholder' label. Below these is a 'Message' section with a 'Type here' label and a large text area. At the bottom, there is contact information: 'Tel Number' (+27XXXXXX), 'Email Address' (XXXX@examole.com), and 'Physical Address' (XXXXXXXXXXXXXXXXXX). A map placeholder is shown below the address, and a copyright notice 'Copyright © 2020' is at the very bottom.

Figure 11 shows the wireframe of the Contact Page

The role of the contact page is to display the necessary contact information such as the telephone number, email address and physical address of the company. As such, our contact page will follow this standard, but we will make sure that the contact information will have dynamic call-to-action features. This allows the user to click/tap on the telephone number, email address or physical address, and will immediately be able to call, mail or navigate with a quick click/tap of a button.

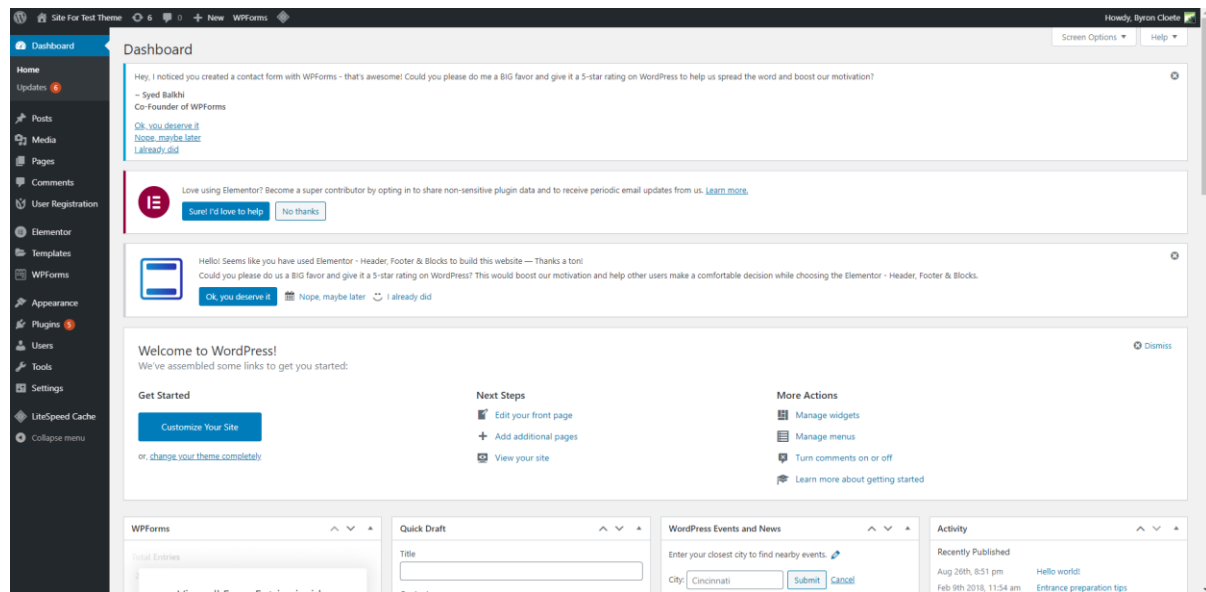
Design Tools

WordPress

Elementor - frontend drag & drop page builder

Design Tools

WordPress



WordPress is a free and open-source content management system written in PHP and paired with a MySQL or MariaDB database. Features include a plugin architecture and a template system, referred to within WordPress as Themes ('Wikipedia', 2020).

This platform allows us easily create a website that has access to a vast selection customisation tools that will consistently provide an answer for our design problems. The WordPress platform is user-friendly, even for beginners like us and is well-supported, so we will always find discussions online that tackle problems we might run into. WordPress allows for the usage of themes, which makes it possible to fine-tune the appearance and functionality of our website. Another positive is that there are many free plugins that are supported by the WordPress community and extend the functionality of what is possible with a WordPress website. The sites have built-in mobile responsiveness which make up a large portion of the online userbase.

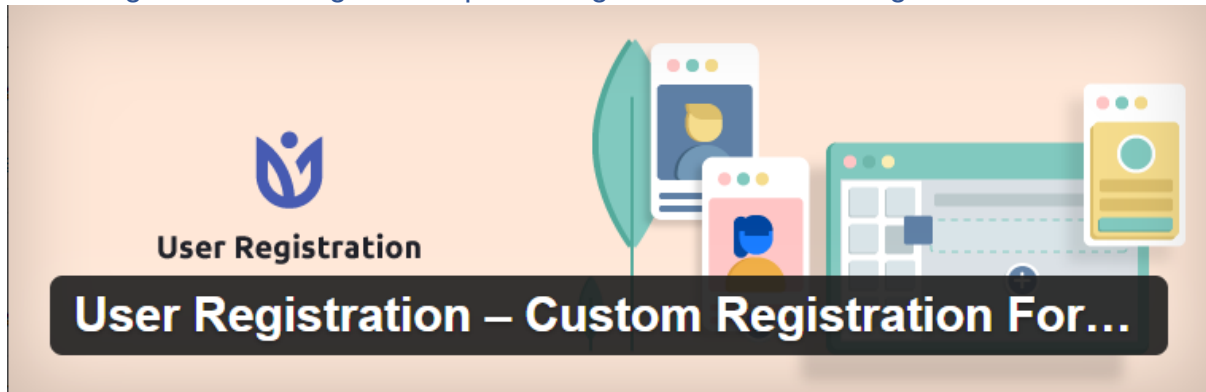
Elementor - frontend drag & drop page builder



The Elementor Website builder allows WordPress users to create and edit websites by employing the drag and drop technique, with a built-in responsive mode ('Elementor', 2020).

We will be using this tool to create our WordPress website. It comes with a host of templates that are easy to use and offer drag and drop functionality. This is the design recommended by WordPress professionals and will serve the group well in the creation of our website. Though the tool comes with a premium/paid service, we will be using the free version because it has sufficient features. The plugins we want to use with this page builder work well and won't suffer any conflicts.

User Registration - drag and drop user registration form and login form builder



User Registration plugin provides you with an easy way to create frontend user registration form and login form. Drag and Drop fields make ordering and creating forms extremely easy. The plugin is lightweight, extendible, and can be used to create any type of registration form.

Beside registration form, the plugin also supports beautiful frontend profile account page for profile edit, password change, Log out and more. Users can visit their account page after registration and view the details they have filled and can make changes if necessary.

The registration forms are 100% mobile responsive and optimized to display on any devices.

Timeline

Project Members :				Q3 2020					Q4 2020	
Mr Cloete	Ms. Irakoze	Mr Kokolo	Mr Villet	Jun 2020	Jul 2020	Aug 2020	Sep 2020	Oct 2020	Nov 2020	Project 2020
Ms. May	Mr Ruiters	Mr Villet	Mr Ngodivane							
Mr De-Tchambila										
Assignment Tracker										
1										
2	Initiation Phase									
3	Brainstorming Ideas			06/02/20	06/05/20	All members				
4	Decide Project Topic			06/05/20	06/07/20	All members				
5	Problem Statement and Proposed Solution			06/07/20	06/12/20	Mr De-Tchambila Ms. May				
6	Planning Phase									
7	Identify Entity/Role/Relationships			06/12/20	06/16/20	Mr Villet				
8	UML Domain Diagram Design			06/16/20	06/20/20	All members				
9	WordPress			07/21/20	11/08/20	All members				WordPress
10	UI description			09/04/20	09/24/20	All members				UI description
11	Detailed design			09/11/20	09/24/20	All members				Detailed design
12	Test design Specifications			09/22/20	09/24/20	All members				Test design Specifications
13	Implementation Phase									
14	Find Extra Needed Resources			09/25/20	09/28/20	All members				Find Extra Needed Resources
15	Work on assigned deliverables			09/25/20	11/08/20	All members				Work on assigned deliverables
16	Monitoring and control processes			09/25/20	11/13/20	Mr Cloete				Monitoring and control processes
17	Closure Phase									
18	Releasing the final deliverables			11/20/20	11/20/20	Mr Cloete				Releasing the final deliverables
19	Post-mortem meeting			11/21/20	11/21/20	All members				Post-mortem meeting

Testing

Home Page Testing

To test the homepage, we will check whether users can access the services of the website without creating an account.

Certain links should not show for the user that is not registered, whether it be in the navigation bar, or in the content of the website.

Once the user has created their profile the content of the homepage should change as well as the elements in the navigation bar, the user should gain access to the services they were previously denied.

We will test whether the information displayed on the home page matches the information associated with the user.

User Registration/Login Testing

If the user does not have an account then they should not be able to login.

If the user already has an account associated with either their email or student number, they should not be able to create a new account.

The text boxes should all be filled in with the relevant information, if not then the user should be prompted to input the relevant information.

If the text that has been inserted is in the incorrect format, i.e. the email address (xxxxx@gmail.com) or the student number (214189872) then the user should be prompted to input the correct formatting.

Assignment Creation Testing

To test the creation of an assignment a few test cases are considered:

- when a user does not have an account, they are not permitted to create any assignment
- once a user is registered, they are then permitted to create an assignment with the required details for the assignment.
- Every assignment created must be associated with an existing subject and lecturer
- if an assignment is individual it will not allow for polls to be run
- an assignment cannot be created with a deadline in the past. the system must prompt the user to set the deadline date the same day or a day in the future.
- an assignment cannot be created without a deadline or an assigned date. if the assigned date is omitted the system automatically sets the assigned date to the current date.
- During the creation of group assignments, the creator is automatically set as the administrator of the assignment.
- Group assignments are required to have more than one contributor.
Therefore, the creator is automatically prompted to invite at least another user to the assignment before any contribution made on the assignment. Failure to do so, the system set the created assignment as a group assignment until more members are added to the assignment.

Add Subject Testing

Test Case: Add subject with empty/filled data fields

Requirement: The user attempts to press the add button when one or all fields are empty.

1. Provide empty subject in the subject name textbox or/and
2. Provide empty subject in the subject name textbox or/and
3. Choose empty lecturer in the list of lecturers or/and
4. Press the "Add" button.

Expected result: The subject is not added. The system provides a message requesting users to enter subject or/and subject codes fields.

Contact Page Testing

Firstly, we'll create or use one of our existing contact numbers/email addresses. These hyperlinks and buttons use a pop-up prompt for the user to engage with.

- To test the phone number, we will use one of our numbers and use the call feature via both the "call now" button and hyperlink on mobile (calling via a desktop browser won't work)
- To test the email address and contact form, we will use one of our student email addresses via the email hyperlink and contact form from the desktop browser and mobile browser.
- To test the physical address and map, we'll use the university's physical and the user will be prompted to select one of the navigation apps to display and navigate to the address.

Lecturer Page Testing

We will test the lecturer page to check first if the user is able to access the page through the home page

To check if the lecturer is able to create an account, make sure he filled all the required information, before validating the account. For example if he didn't put his name the system must display the message with the correct errors stated.

Check that a user who doesn't have an account is not able to post an assignment. And make sure that the page is mobile friendly by using different devices such as mobile phone and tablet.

Bibliography

'WordPress' (2020). *Wikipedia*. Available at <https://en.wikipedia.org/wiki/WordPress> (Accessed: 25 September 2020)

'Elementor' (2020). *Wikipedia*. Available at <https://en.wikipedia.org/wiki/Elementor> (Accessed: 25 September 2020)